

The pain sensation immediately ceases when the individual moves out of the beam or when the beam is turned off. The millimetre-wave beam does not cause any injury because of the low energy levels used and the shallow penetration depth of about 0.5mm at the operating wavelength.

In addition, pain induced as a consequence of the subject's natural defence mechanism serves as a warning to help protect him or her from injury. A simple control console on ADS enables the operator to view the scene and precisely aim



Fig. 10: Vigilant Eagle protection system concept

the millimetre-wave beam to affect only the intended target.

ADS has been comprehensively evaluated on a large number of volunteers to establish its non-lethality. Two variants of ADS integrated on land vehicles include ADS mounted on a wheeled vehicular platform and another integrated on high-mobility, multi-purpose wheeled vehicle platform more commonly known as Humvee (Fig. 8). Such platforms would be equipped with required power source to drive the weapon system.

A mid-range version of ADS called Silent Guardian with an operational range of about 250 metres has also been developed by Raytheon (Fig. 9). Silent Guardian is primarily developed for use by law-enforcement agencies. The system is easily transportable on standard

military tactical vehicles and can also be integrated into combat vehicles.

It is also being considered for operation from airborne platforms such as AC-130 gunship. Airborne version will be a more powerful and lightweight version of its land based counterpart. The USA reportedly deployed ADS as a crowd-control weapon in Afghanistan, but it was withdrawn in 2010 without ever seeing it being used due to serious questions raised by critics about the ethics of using a pain beam to break up riots.

According to reports, Russia and China are developing their own versions of ADS. As reported by Interfax News Agency, Russian military is testing its own beam weapon under development at 12th Central Military Research and Development Institute. The device reportedly has a range of about 270 metres and is intended to be used for riot-control applications.

The Chinese system called Poly WB-1 pain beam weapon also uses millimetre-wave energy to cause intolerable pain in human subjects up to a kilometre away from the source.

Vigilant Eagle. This HPM weapon system from Raytheon is designed to protect both commercial flights as well as military and private aircraft from the threat of surface-to-air missiles including Man Portable Air Defence Systems (MANPADS). The system is installed at airports or airfields and is primarily intended to provide protection to the aircraft while taking off from or landing at the airports or airfields by directing a precisely-steered beam of electromagnetic energy, a set of high-frequency microwaves, at the missile threatening the aircraft in

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Fax : 0091-22-27873310, 27673330

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